THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND THREE CHAINLINK SOLUTIONS

Sponsored by


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1 Set 3/Game 1

1. $\displaystyle\int_1^4 \{x\}^{\lfloor x \rfloor}\ \mathrm{d}x$

   Solution.

   $$\int_1^4 \{x\}^{\lfloor x \rfloor}\ \mathrm{d}x = \sum_{n=1}^{3} \int_n^{n+1} \{x\}^n\ \mathrm{d}x = \sum_{n=1}^{3} \int_0^1 x^n\ \mathrm{d}x = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12}.$$

   :)

2. $\displaystyle\int \frac{\sinh(x) + \cosh(x)}{\sinh(e^x) + \cosh(e^x)}\ \mathrm{d}x$

   Solution. Converting into exponential form we have

   $$\int \frac{\sinh(x) + \cosh(x)}{\sinh(e^x) + \cosh(e^x)}\ \mathrm{d}x = \int \frac{e^x}{e^{e^x}}\ \mathrm{d}x = -e^{-e^x} + C.$$

   :)

3. $\displaystyle\int \frac{1 + \ln(x)}{\sqrt{x^{-2x} - 1}}\ \mathrm{d}x$

   Solution.

   $$\int \frac{1 + \ln(x)}{\sqrt{x^{-2x} - 1}}\ \mathrm{d}x = \int \frac{x^x(1 + \ln(x))}{\sqrt{1 - x^{2x}}}\ \mathrm{d}x = \sin^{-1}(x^x) + C.$$

   Alternative form: $-\arctan\left(\sqrt{x^{(-2x)} - 1}\right) + C$:)

4. $\displaystyle\int_1^\infty \frac{1}{x\sqrt{x}\sqrt{x-1}}\ \mathrm{d}x$

   Solution. The idea is to choose the right substitutions. The first is $u = \sqrt{x}$. With this we have

   $$\int_1^\infty \frac{1}{x\sqrt{x}\sqrt{x-1}}\ \mathrm{d}x = \int_1^\infty \frac{2}{u^2\sqrt{u^2-1}}\ \mathrm{d}u.$$

   Here, we can pick a few trigonometric substitutions, but the one we choose is $u = \sec(t)$, because the domain of integration is $(1, \infty)$:

   $$\begin{aligned}\int_1^\infty \frac{2}{u^2\sqrt{u^2-1}}\ \mathrm{d}u &= \int_0^{\frac{\pi}{2}} \frac{2}{\sec^2(t)\tan(t)} \cdot \sec(t)\tan(t)\ \mathrm{d}t \\ &= \int_0^{\frac{\pi}{2}} 2\cos(t)\ \mathrm{d}t \\ &= 2.\end{aligned}$$

   :)

5. $\displaystyle\int_0^1 \frac{\ln(1 + x + x^2)}{x}\ \mathrm{d}x$

Solution. We have

$$\begin{aligned}\int_0^1 \frac{\ln(1+x+x^2)}{x}\,\mathrm{d}x &= \int_0^1 \frac{\ln(1+x+x^2)+\ln(1-x)-\ln(1-x)}{x}\,\mathrm{d}x\\ &= \int_0^1 \frac{\ln(1-x^3)-\ln(1-x)}{x}\,\mathrm{d}x\\ &= -\sum_{n=1}^{\infty}\int_0^1 \frac{x^{3n-1}}{n}-\frac{x^{n-1}}{n}\,\mathrm{d}x\\ &= \frac{2}{3}\sum_{n=1}^{\infty}\frac{1}{n^2}\\ &= \frac{\pi^2}{9}.\end{aligned}$$

:)

6. $\displaystyle\int_0^1 f(x)\,\mathrm{d}x$, where $f(x)=\begin{cases}1, & x=0.x_1x_2x_3\ldots \text{ where } x_n\neq 9 \text{ for all } n;\\ 0, & \text{else}\end{cases}$

Solution. Let $A=\{x : f(x)=1\}$. A is a Cantor set formed by

- in the first step removing 9^0 intervals of Lebesgue measure $\dfrac{1}{10^1}$,
- in the second step removing 9^1 intervals of Lebesgue measure $\dfrac{1}{10^2}$,
- in general at the nth step removing 9^{n-1} intervals of measure $\dfrac{1}{10^n}$.

Hence the Lebesgue measure of A is

$$1-\frac{1}{10}\sum_{n=0}^{\infty}\frac{9^n}{10^n}=1-\frac{1}{10}\cdot\frac{1}{1-\frac{9}{10}}=0.$$

It follows $\displaystyle\int_0^1 f(x)\,\mathrm{d}x=0$. :)

7. $\displaystyle\int_{\tan^{-1}\left(\frac{1}{3}\right)}^{\tan^{-1}\left(\frac{1}{2}\right)} \frac{x}{\sin(4x)}\,\mathrm{d}x$

Solution. By inverse tangent addition formula,

$$\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{2}\right)=\tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{2}}{1-\frac{1}{3}\cdot\frac{1}{2}}\right)=\tan^{-1}(1)=\frac{\pi}{4}.$$

Hence we write

$$\begin{aligned}\int_{\tan^{-1}\left(\frac{1}{3}\right)}^{\tan^{-1}\left(\frac{1}{2}\right)} \frac{x}{\sin(4x)}\,\mathrm{d}x &= \frac{1}{16}\int_{4\tan^{-1}\left(\frac{1}{3}\right)}^{4\tan^{-1}\left(\frac{1}{2}\right)} \frac{y}{\sin(y)}\,\mathrm{d}y\\ &= \frac{1}{16}\int_{4\tan^{-1}\left(\frac{1}{3}\right)}^{4\tan^{-1}\left(\frac{1}{2}\right)} \frac{\pi-y}{\sin(\pi-y)}\,\mathrm{d}y\\ &= \frac{1}{16}\int_{4\tan^{-1}\left(\frac{1}{3}\right)}^{4\tan^{-1}\left(\frac{1}{2}\right)} \frac{\pi-y}{\sin(y)}\,\mathrm{d}y.\end{aligned}$$

Hence we get

$$\begin{aligned}
&\int_{\tan^{-1}\left(\frac{1}{3}\right)}^{\tan^{-1}\left(\frac{1}{2}\right)} \frac{x}{\sin(4x)}\,\mathrm{d}x \\
&= \frac{\pi}{32}\int_{4\tan^{-1}\left(\frac{1}{3}\right)}^{4\tan^{-1}\left(\frac{1}{2}\right)} \frac{1}{\sin(x)}\,\mathrm{d}x \\
&= \frac{\pi}{32}\left[\ln\left|\csc(x)-\cot(x)\right|\right]_{4\tan^{-1}\left(\frac{1}{3}\right)}^{4\tan^{-1}\left(\frac{1}{2}\right)} \\
&= \frac{\pi}{32}\left(\ln\left|\csc\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)-\cot\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)\right|-\ln\left|\csc\left(4\tan^{-1}\left(\frac{1}{3}\right)\right)-\cot\left(4\tan^{-1}\left(\frac{1}{3}\right)\right)\right.\right.
\end{aligned}$$

It's fine to leave it here, but with a bit of work, this can be significantly simplified. Observing once more that $4\tan^{-1}\left(\frac{1}{2}\right)+4\tan^{-1}\left(\frac{1}{3}\right)=\pi$, we have

$$\begin{aligned}
\csc\left(4\tan^{-1}\left(\frac{1}{3}\right)\right) &= \csc\left(\pi-4\tan^{-1}\left(\frac{1}{2}\right)\right) \\
&= \csc\left(4\tan^{-1}\left(\frac{1}{2}\right)\right), \\
\cot\left(4\tan^{-1}\left(\frac{1}{3}\right)\right) &= \cot\left(\pi-4\tan^{-1}\left(\frac{1}{2}\right)\right) \\
&= -\cot\left(4\tan^{-1}\left(\frac{1}{2}\right)\right),
\end{aligned}$$

so that the value becomes

$$\begin{aligned}
&\frac{\pi}{32}\ln\left|\frac{\csc\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)-\cot\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)}{\csc\left(4\tan^{-1}\left(\frac{1}{3}\right)\right)-\cot\left(4\tan^{-1}\left(\frac{1}{3}\right)\right)}\right| \\
&= \frac{\pi}{32}\ln\left|\frac{\csc\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)-\cot\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)}{\csc\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)+\cot\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)}\right| \\
&= \frac{\pi}{32}\ln\left(\frac{1-\cos\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)}{1+\cos\left(4\tan^{-1}\left(\frac{1}{2}\right)\right)}\right).
\end{aligned}$$

Now, cosine double angle twice gives

$$\cos(4x) = 2(2\cos^2(x)-1)^2-1 = 8\cos^4(x)-8\cos^2(x)+1.$$

With

$$\cos\left(\tan^{-1}\left(\frac{1}{2}\right)\right) = \frac{2}{\sqrt{5}},$$

we get

$$\cos\left(4\tan^{-1}\left(\frac{1}{2}\right)\right) = 8\cdot\frac{16}{25}-8\cdot\frac{4}{5}+1 = -\frac{7}{25}.$$

Thus the required value is

$$\frac{\pi}{32}\ln\left(\frac{1+\frac{7}{25}}{1-\frac{7}{25}}\right) = \frac{\pi}{32}\ln\left(\frac{16}{9}\right) = \frac{\pi}{16}\ln\left(\frac{4}{3}\right).$$

:)

8. $\lim_{n\to\infty}(n+1)\int_0^1 x^{x^x+n}\ln(1+x+x^x+x^{x^x}+x^{x^{x^x}})\,\mathrm{d}x$

Solution. More generally consider for $f:[0,1]\to\mathbb{R}$ integrable and continuous at 1, the limit

$$\lim_{n\to\infty}(n+1)\int_0^1 x^n f(x)\,\mathrm{d}x.$$

Claim this is $f(1)$. We have

$$\begin{aligned}\left|(n+1)\int_0^1 x^n f(x)\,\mathrm{d}x - f(1)\right| &= \left|(n+1)\int_0^1 x^n(f(x)-f(1))\,\mathrm{d}x\right| \\ &\le (n+1)\int_0^1 x^n|f(x)-f(1)|\,\mathrm{d}x.\end{aligned}$$

It turns out we can bound this arbitrarily small. Let $\varepsilon>0$ be given. Then there exists $0<\delta<1$ such that $|f(x)-f(1)|<\dfrac{\varepsilon}{2}$ on $(1-\delta,1]$. For n sufficiently large we have $(n+1)(1-\delta)^n<\dfrac{\varepsilon}{2(\int_0^1|f(x)|\,\mathrm{d}x+|f(1)|)}$. Then it follows

$$\begin{aligned}&(n+1)\int_0^1 x^n|f(x)-f(1)|\,\mathrm{d}x \\ &\quad=(n+1)\int_0^{1-\delta} x^n|f(x)-f(1)|\,\mathrm{d}x+(n+1)\int_{1-\delta}^1 x^n|f(x)-f(1)|\,\mathrm{d}x \\ &\quad<(n+1)(1-\delta)^n\int_0^1|f(x)|+|f(1)|\,\mathrm{d}x+\frac{\varepsilon}{2}(n+1)\int_0^1 x^n\,\mathrm{d}x \\ &\quad<\varepsilon.\end{aligned}$$

This proves the claim. Then with $f(x)=x^{x^x}\ln(1+x+x^x+x^{x^x}+x^{x^{x^x}})$, we get $f(1)=\ln(5)$. :)

2 Set 5/Game 2

1. $\displaystyle\int_{-1}^{0} 5(x^6+x)^4 \,\mathrm{d}x$

   Solution.

$$\int_{-1}^{0} 5(x^6+x)^4 \,\mathrm{d}x = \int_{-1}^{0} 5x^4(x^5+1)^4 \,\mathrm{d}x = \left[\frac{(x^5+1)^5}{5}\right]_{-1}^{0} = \frac{1}{5}.$$

   :)

2. $\displaystyle\int e^{e^{e^x}} e^{e^x} e^{e^x} e^x \,\mathrm{d}x$

   Solution. Let $u = e^{e^x}$. Then $u' = e^{e^x} e^x$, and the integral is

$$\int e^{e^{e^x}} e^{e^x} e^{e^x} e^x \,\mathrm{d}x = \int e^u u \,\mathrm{d}u = ue^u - e^u + C = e^{e^{e^x}} \left(e^{e^x} - 1\right) + C.$$

   :)

3. $\displaystyle\int_0^{\frac{\pi}{2}} \sec(x)\csc(x)\sin(\tan(x)) \,\mathrm{d}x$

   Solution. Let $u = \tan(x)$. Then we have

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sec(x)\csc(x)\sin(\tan(x)) \,\mathrm{d}x &= \int_0^{\infty} \sqrt{1+u^2} \cdot \frac{\sqrt{1+u^2}}{u} \cdot \sin(u) \cdot \frac{1}{1+u^2} \,\mathrm{d}u \\ &= \int_0^{\infty} \frac{\sin(u)}{u} \,\mathrm{d}u \\ &= \frac{\pi}{2}.\end{aligned}$$

   :)

4. $\displaystyle\int_0^{\infty} \ln(1+e^{-x})\cos(x) \,\mathrm{d}x$

   Solution.

$$\begin{aligned}I &= \int_0^{\infty} \ln\left(1+e^{-x}\right)\cos(x) \,\mathrm{d}x \\ &= \int_0^{\infty} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}e^{-kx}}{k} \cos(x) \,\mathrm{d}x \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \int_0^{\infty} e^{-kx}\cos(x) \,\mathrm{d}x \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \frac{k}{k^2+1} \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^2+1} \\ &= \frac{1}{2} - \frac{\pi}{2}\operatorname{csch}(\pi)\end{aligned}$$

   :)

5. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\sin(x)}{\cos(x)+\tan(x)}\,\mathrm{d}x$

Solution. We convert everything into sin and cos:

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \frac{\sin(x)}{\cos(x)+\tan(x)}\,\mathrm{d}x &= \int_0^{\frac{\pi}{2}} \frac{\sin(x)\cos(x)}{\cos^2(x)+\sin(x)}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{\sin(x)\cos(x)}{1-\sin^2(x)+\sin(x)}\,\mathrm{d}x.\end{aligned}$$

Let $u=\sin(x)$, then the integral becomes

$$\begin{aligned}\int_0^1 \frac{u}{1+u-u^2}\,\mathrm{d}u &= -\frac{1}{2}\int_0^1 \frac{2u}{u^2-u-1}\,\mathrm{d}u \\ &= -\int_0^1 \frac{2u-1}{u^2-u-1}+\frac{1}{\left(u-\frac{1}{2}\right)^2-\frac{5}{4}}\,\mathrm{d}u \\ &= -\left[\ln|u^2-u-1|-\frac{2}{\sqrt{5}}\tanh^{-1}\left(\frac{2u-1}{\sqrt{5}}\right)\right]_0^1 \\ &= \frac{2}{\sqrt{5}}\tanh^{-1}\left(\frac{1}{\sqrt{5}}\right).\end{aligned}$$

:)

6. $\displaystyle\int_0^{\infty} e^{-a^2x^2-\frac{b^2}{x^2}}\,\mathrm{d}x$ for $a,b>0$

Solution. Let $\sqrt{\frac{b}{a}}u=x$. Then we have

$$\begin{aligned}\int_0^{\infty} e^{-a^2x^2-\frac{b^2}{x^2}}\,\mathrm{d}x &= \sqrt{\frac{b}{a}}\int_0^{\infty} e^{-abu^2-\frac{ab}{u^2}}\,\mathrm{d}u \\ &= \frac{1}{2}\sqrt{\frac{b}{a}}\int_{-\infty}^{\infty} e^{-abu^2-\frac{ab}{u^2}}\,\mathrm{d}u \\ &= \frac{1}{2e^{2ab}}\sqrt{\frac{b}{a}}\int_{-\infty}^{\infty} e^{-abu^2}\,\mathrm{d}u \\ &= \frac{1}{2e^{2ab}}\sqrt{\frac{b}{a}}\sqrt{\frac{\pi}{ab}} \\ &= \frac{\sqrt{\pi}}{2ae^{2ab}}.\end{aligned}$$

:)

7. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\cos^3(x)\sin(x)}{\ln(\sin(x))}\,\mathrm{d}x$

Solution. Let $u=\sin(x)$ and get

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \frac{\cos^3(x)\sin(x)}{\ln(\sin(x))}\,\mathrm{d}x &= \int_0^{\frac{\pi}{2}} \frac{(1-\sin^2(x))\cos(x)\sin(x)}{\ln(\sin(x))}\,\mathrm{d}x \\ &= \int_0^1 \frac{(1-u^2)u}{\ln(u)}\,\mathrm{d}u.\end{aligned}$$

Further let $y = u^2$. Then we get

$$\int_0^1 \frac{(1-u^2)u}{\ln(u)}\,\mathrm{d}u = \int_0^1 \frac{1-y}{\ln(y)}\,\mathrm{d}y.$$

This last integral is evaluated using DUTIS. Let $I(a) = \displaystyle\int_0^1 \frac{1-y^a}{\ln(y)}\,\mathrm{d}y$, then

$$I'(a) = -\int_0^1 y^a\,\mathrm{d}y = -\frac{1}{a+1}.$$

Hence $I(a) = -\ln|a+1| + C$, and putting $a = 0$ gives $C = 0$. Hence it follows $I(1) = -\ln(2)$, which was the value needed. :)

8. $\displaystyle\lim_{n\to\infty}\int_0^1 2025\sin(\{nx^{2025}\})\cos(\{nx^{2025}\})\,\mathrm{d}x$

Solution. We show a more general statement, that

$$\lim_{n\to\infty}\int_0^1 f(\{nx^k\})\,\mathrm{d}x = \int_0^1 f(x)\,\mathrm{d}x.$$

As usual, we split the bounds as

$$\begin{aligned}\int_0^1 f(\{nx^k\})\,\mathrm{d}x &= \sum_{m=0}^{n-1}\int_{\left(\frac{m}{n}\right)^{\frac{1}{k}}}^{\left(\frac{m+1}{n}\right)^{\frac{1}{k}}} f(nx^k-m)\,\mathrm{d}x\\ &= \sum_{m=0}^{n-1}\int_0^1 \frac{1}{nk}\left(\frac{m+t}{n}\right)^{\frac{1}{k}-1} f(t)\,\mathrm{d}t\\ &= \int_0^1 f(t)\sum_{m=0}^{n-1}\frac{1}{kn}\left(\frac{m+t}{n}\right)^{\frac{1}{k}-1}\mathrm{d}t\end{aligned}$$

where in the second line we have substituted $t = nx^k - m$.

But observe that for fixed $t \in (0,1)$ we have the Riemann sum

$$\lim_{n\to\infty}\sum_{m=0}^{n-1}\frac{1}{kn}\left(\frac{m+t}{n}\right)^{\frac{1}{k}-1} = \int_0^1 \frac{1}{k}u^{\frac{1}{k}-1}\,\mathrm{d}u = 1.$$

It follows from interchange of limit and integral (formally, can be justified by dominated convergence) that

$$\lim_{n\to\infty}\int_0^1 f(t)\sum_{m=0}^{n-1}\frac{1}{kn}\left(\frac{m+t}{n}\right)^{\frac{1}{k}-1}\mathrm{d}t = \int_0^1 f(t)\,\mathrm{d}t.$$

Now in the problem, $f(x) = 2025\sin(x)\cos(x)$, so the required value is

$$\int_0^1 2025\sin(x)\cos(x)\,\mathrm{d}x = \frac{2025}{2}\sin^2(1).$$

:)

3 Set 4/Game 3

1. $\displaystyle\int_0^\infty e^{-x^{\frac{1}{2025}}}\,\mathrm{d}x$

   Solution. Let $y = x^{\frac{1}{2025}}$. It follows

   $$\int_0^\infty e^{-x^{\frac{1}{2025}}}\,\mathrm{d}x = 2025\int_0^\infty y^{2024}e^{-y}\,\mathrm{d}x = 2025!.$$

   :)

2. $\displaystyle\int_{-1}^\infty e^{-x^2}(5x^4 - 2x^6)\,\mathrm{d}x$

   Solution.

   $$\int_{-1}^\infty e^{-x^2}(5x^4 - 2x^6)\,\mathrm{d}x = \int_{-1}^\infty e^{-x^2}\cdot 5x^4 + (-2xe^{-x^2})\cdot x^5\,\mathrm{d}x = [x^5e^{-x^2}]_{-1}^\infty = \frac{1}{e}.$$

   :)

3. $\displaystyle\int (x + 2x^2 + 3x^3 + \cdots + 2025x^{2025})(1 + x + x^2 + \cdots + x^{2024})\,\mathrm{d}x$

   Solution.

   $$\begin{aligned}
   &\int (x + 2x^2 + 3x^3 + \cdots + 2025x^{2025})(1 + x + x^2 + \cdots + x^{2024})\,\mathrm{d}x \\
   &= \int (1 + 2x + 3x^2 + \cdots + 2025x^{2024})(x + x^2 + x^3 + \cdots + x^{2025})\,\mathrm{d}x \\
   &= \frac{1}{2}(x + x^2 + x^3 + \cdots + x^{2025})^2 + C \\
   &= \frac{1}{2}\left(\frac{x - x^{2026}}{1 - x}\right)^2 + C.
   \end{aligned}$$

   :)

4. $\displaystyle\int_0^{\frac{\pi}{4}} \frac{\sin(2x)}{\sin(x) + \cos(x) + 1}\,\mathrm{d}x$

   Solution.

   $$\begin{aligned}
   \int_0^{\frac{\pi}{4}} \frac{\sin(2x)}{\sin(x) + \cos(x) + 1}\,\mathrm{d}x &= \int_0^{\frac{\pi}{4}} \frac{2\sin(x)\cos(x)}{\sin(x) + \cos(x) + 1}\,\mathrm{d}x \\
   &= \int_0^{\frac{\pi}{4}} \frac{(\sin(x) + \cos(x))^2 - 1}{\sin(x) + \cos(x) + 1}\,\mathrm{d}x \\
   &= \int_0^{\frac{\pi}{4}} \sin(x) + \cos(x) - 1\,\mathrm{d}x \\
   &= [\sin(x) - \cos(x) - x]_0^{\frac{\pi}{4}} \\
   &= 1 - \frac{\pi}{4}.
   \end{aligned}$$

   :)

5. $\displaystyle\int_0^{\frac{\pi}{2}} \ln(\sin(x))\tan(x)\,\mathrm{d}x$

Solution. Let $u = \cos(x)$. Then we have

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \ln(\sin(x)) \tan(x)\, \mathrm{d}x &= \int_0^1 \frac{\ln(\sqrt{1-u^2})}{u}\, \mathrm{d}u \\ &= \frac{1}{2}\int_0^1 \frac{\ln(1-u^2)}{u}\, \mathrm{d}u \\ &= -\frac{1}{2}\sum_{n=1}^{\infty}\int_0^1 \frac{u^{2n-1}}{n}\, \mathrm{d}u \\ &= -\frac{1}{4}\sum_{n=1}^{\infty}\frac{1}{n^2}\, \mathrm{d}u \\ &= -\frac{\pi^2}{24}\end{aligned}$$

:)

6. $\displaystyle\int_0^{\infty} \frac{x^{\varphi}\tan^{-1}(x)}{1+x^{2\varphi^2}}\, \mathrm{d}x$, where $\varphi = \dfrac{1+\sqrt{5}}{2}$ is the golden ratio

Solution. Using $\varphi^2 = \varphi + 1$ and $u = x^{\varphi+1}$, we have

$$\begin{aligned}I &:= \int_0^{\infty} \frac{x^{\varphi}\tan^{-1}(x)}{1+x^{2+2\varphi}}\, \mathrm{d}x \\ &= \frac{1}{\varphi+1}\int_0^{\infty} \frac{\tan^{-1}\left(u^{\frac{1}{\varphi+1}}\right)}{1+u^2}\, \mathrm{d}u \\ &= \frac{1}{\varphi+1}\int_0^{\infty} \frac{\tan^{-1}\left(v^{-\frac{1}{\varphi+1}}\right)}{1+v^2}\, \mathrm{d}v,\end{aligned}$$

where in the last line we have performed a border flip $u = \dfrac{1}{v}$ substitution. It follows now from the identity $\tan^{-1}(x) + \tan^{-1}\left(x^{-1}\right) = \dfrac{\pi}{2}$ that

$$\begin{aligned}I &= \frac{1}{2(\varphi+1)}\int_0^{\infty} \frac{\tan^{-1}\left(u^{\frac{1}{\varphi+1}}\right) + \tan^{-1}\left(u^{-\frac{1}{\varphi+1}}\right)}{1+u^2}\, \mathrm{d}u \\ &= \frac{\pi}{4(\varphi+1)}\int_0^{\infty} \frac{1}{1+u^2}\, \mathrm{d}u \\ &= \frac{\pi^2}{8(\varphi+1)} \\ &= \frac{\pi^2}{8\varphi^2}.\end{aligned}$$

:)

7. $\displaystyle\int_0^2 (3x^2 - 5x + 3)\cos(x^3 - 3x^2 + 4x - 2)\, \mathrm{d}x$

Solution. Let $x = u + 1$. Then we have

$$\begin{aligned}
&\int_0^2 (3x^2 - 5x + 3)\cos(x^3 - 3x^2 + 4x - 2)\,\mathrm{d}x \\
&= \int_{-1}^1 (3(u+1)^2 - 5(u+1) + 3)\cos((u+1)^3 - 3(u+1)^2 + 4(u+1) - 2)\,\mathrm{d}u \\
&= \int_{-1}^1 (3u^2 + u + 1)\cos(u^3 + u)\,\mathrm{d}u \\
&= \int_{-1}^1 (3u^2 + 1)\cos(u^3 + u)\,\mathrm{d}u + \int_{-1}^1 u\cos(u^3 + u)\,\mathrm{d}u \\
&= [\sin(u^3 + u)]_{-1}^1 + 0 \\
&= 2\sin(2).
\end{aligned}$$

:)

8. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{1 - \ln(\cos(x))}{x^2 + (1 - \ln(\cos(x)))^2}\cos(2x)\,\mathrm{d}x$

Solution. It sounds crazy, but the first step is to convert it into a double integral using a Laplace transform:

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \frac{1 - \ln(\cos(x))}{x^2 + (1 - \ln(\cos(x)))^2}\cos(2x)\,\mathrm{d}x &= \int_0^{\frac{\pi}{2}} \cos(2x)\int_0^\infty \cos(t)e^{-(1-\ln(\cos(x)))t}\,\mathrm{d}t\,\mathrm{d}x \\
&= \int_0^{\frac{\pi}{2}}\int_0^\infty \cos(2x)\cos^t(x)\cos(xt)e^{-t}\,\mathrm{d}t\,\mathrm{d}x \\
&= \int_0^\infty e^{-t}\int_0^{\frac{\pi}{2}} \cos(2x)\cos^t(x)\cos(xt)\,\mathrm{d}x\,\mathrm{d}t.
\end{aligned}$$

The inner integral is a goated exercise in tricks. First, we use evenness (for a later step) and complexify:

$$\int_0^{\frac{\pi}{2}} \cos(2x)\cos^t(x)\cos(xt)\,\mathrm{d}x = \frac{1}{2}\,\mathrm{Re}\left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(2x)\cos^t(x)e^{ixt}\,\mathrm{d}x\right).$$

Observe that $2\cos(x)e^{ix} = e^{2ix} + 1$. Indeed, from $2\cos(x) = e^{ix} + e^{-ix}$ this is evident.

And now for tricks galore. The RHS is

$$\begin{aligned}
&\frac{1}{2^{t+1}}\operatorname{Re}\left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(e^{2ix}+1)^t\cos(2x)\,\mathrm{d}x\right)\\
&\quad=\frac{1}{2^{t+2}}\operatorname{Re}\left(\int_{-\pi}^{\pi}(e^{ix}+1)^t\cos(x)\,\mathrm{d}x\right)\\
&\quad=\lim_{r\to1^-}\frac{1}{2^{t+2}}\operatorname{Re}\left(\int_{-\pi}^{\pi}(re^{ix}+1)^t\cos(x)\,\mathrm{d}x\right)\\
&\quad=\lim_{r\to1^-}\frac{1}{2^{t+2}}\operatorname{Re}\left(\int_{-\pi}^{\pi}\sum_{n=0}^{\infty}\binom{t}{n}r^ne^{ixn}\cos(x)\,\mathrm{d}x\right)\\
&\quad=\lim_{r\to1^-}\frac{1}{2^{t+2}}\sum_{n=0}^{\infty}\binom{t}{n}r^n\operatorname{Re}\left(\int_{-\pi}^{\pi}e^{ixn}\cos(x)\,\mathrm{d}x\right)\\
&\quad=\lim_{r\to1^-}\frac{1}{2^{t+3}}\sum_{n=0}^{\infty}\binom{t}{n}r^n\operatorname{Re}\left(\int_{-\pi}^{\pi}e^{ix(n+1)}+e^{ix(n-1)}\,\mathrm{d}x\right)\\
&\quad=\lim_{r\to1^-}\frac{1}{2^{t+3}}\binom{t}{1}r\cdot2\pi\\
&\quad=\frac{t\pi}{2^{t+2}}.
\end{aligned}$$

Finally we evaluate

$$\frac{\pi}{4}\int_0^\infty t(2e)^{-t}\,\mathrm{d}t=\frac{\pi}{4}\int_0^\infty te^{-(1+\ln(2))t}\,\mathrm{d}t=\frac{\pi}{4(1+\ln(2))^2}.$$

:)

4 Set 1/Game 4

1. $\displaystyle\int \frac{\sin^2(x)}{\cos^4(x)}\,\mathrm{d}x$

   Solution.

$$\int \frac{\sin^2(x)}{\cos^4(x)}\,\mathrm{d}x = \int \tan^2(x)\sec^2(x)\,\mathrm{d}x = \frac{\tan^3(x)}{3} + C.$$

:)

2. $\displaystyle\int_0^1 \frac{1+\sqrt{1-x^2}}{\sqrt{1+x}+\sqrt{1-x}}\,\mathrm{d}x$

   Solution. Rationalise the denominator to get

$$\begin{aligned}
\int_0^1 \frac{1+\sqrt{1-x^2}}{\sqrt{1+x}+\sqrt{1-x}}\,\mathrm{d}x &= \int_0^1 \frac{(1+\sqrt{1-x^2})(\sqrt{1+x}-\sqrt{1-x})}{2x}\,\mathrm{d}x \\
&= \int_0^1 \frac{\sqrt{1+x}-\sqrt{1-x}+(1+x)\sqrt{1-x}-(1-x)\sqrt{1+x}}{2x}\,\mathrm{d}x \\
&= \frac{1}{2}\int_0^1 \sqrt{1+x}+\sqrt{1-x}\,\mathrm{d}x \\
&= \frac{1}{3}\left[(1+x)^{\frac{3}{2}}-(1-x)^{\frac{3}{2}}\right]_0^1 \\
&= \frac{2\sqrt{2}}{3}.
\end{aligned}$$

:)

3. $\displaystyle\int_1^\infty \frac{(-1)^{\lceil x\rceil}}{\sqrt{\lfloor x\rfloor\,\lceil x\rceil}\left(\sqrt{\lceil x\rceil}-\sqrt{\lfloor x\rfloor}\right)}\,\mathrm{d}x$

   Solution.

$$\begin{aligned}
\int_1^\infty \frac{(-1)^{\lceil x\rceil}}{\sqrt{\lfloor x\rfloor\,\lceil x\rceil}\left(\sqrt{\lceil x\rceil}-\sqrt{\lfloor x\rfloor}\right)}\,\mathrm{d}x &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n(n+1)}(\sqrt{n+1}-\sqrt{n})} \\
&= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(\sqrt{n+1}+\sqrt{n})}{\sqrt{n(n+1)}} \\
&= \sum_{n=1}^{\infty} (-1)^{n+1}\left(\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{n+1}}\right) \\
&= 1.
\end{aligned}$$

:)

4. $\displaystyle\int \sqrt{\frac{\ln(x+\sqrt{1+x^2})}{1+x^2}}\,\mathrm{d}x$

   Solution. Let $u = \ln(x+\sqrt{1+x^2})$. Then it follows $\mathrm{d}u = \dfrac{1+\frac{2x}{2\sqrt{1+x^2}}}{x+\sqrt{1+x^2}}\,\mathrm{d}x = \dfrac{1}{\sqrt{1+x^2}}\,\mathrm{d}x$.

Hence,

$$\begin{aligned}\int \sqrt{\frac{\ln(x+\sqrt{1+x^2})}{1+x^2}}\,\mathrm{d}x &= \int \sqrt{u}\,\mathrm{d}u \\ &= \frac{2}{3}\left(\ln(x+\sqrt{1+x^2})\right)^{\frac{3}{2}} + C \\ &= \frac{2}{3}\left(\sinh^{-1}(x)\right)^{\frac{3}{2}} + C.\end{aligned}$$

:)

5. $\displaystyle\int_0^{\frac{\pi}{4}} \frac{\cos(2x)}{-\ln(\tan(x))\cos^4(x)}\,\mathrm{d}x$

Solution. Let $u = -\ln(\tan(x))$. Then $x = \tan^{-1}\left(e^{-u}\right)$ and we have $\cos(x) = \dfrac{1}{\sqrt{1+e^{-2u}}}$. Then we get

$$\mathrm{d}x = -\frac{e^{-u}}{1+e^{-2u}}\,\mathrm{d}x$$

and

$$\begin{aligned}\frac{\cos(2x)}{\cos^4(x)} &= \frac{2\cos^2(x)-1}{\cos^4(x)} \\ &= \frac{\frac{2}{1+e^{-2u}}-1}{\frac{1}{(1+e^{-2u})^2}} \\ &= (1+e^{-2u})(1-e^{-2u}).\end{aligned}$$

It follows that

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \frac{\cos(2x)}{-\ln(\tan(x))\cos^4(x)}\,\mathrm{d}x &= \int_0^{\infty} \frac{e^{-u}-e^{-3u}}{u}\,\mathrm{d}u \\ &= \ln(3).\end{aligned}$$

:)

6. $\displaystyle\int \frac{1}{1+x^2}\tan^{-1}\left(\frac{4x-4x^3}{x^4-6x^2+1}\right)\,\mathrm{d}x$

Solution. Let $u = \tan^{-1}\left(\dfrac{4x-4x^3}{x^4-6x^2+1}\right)$. After a tedious calculation we have

$$u' = \frac{4}{1+x^2}.$$

Thus the integral is just

$$\frac{1}{4}\int u\,\mathrm{d}u = \frac{1}{8}\left(\tan^{-1}\left(\frac{4x-4x^3}{x^4-6x^2+1}\right)\right)^2 + C.$$

:)

7. $\lim_{n\to\infty} \int_0^{\frac{1}{2}} f_n(x)\,\mathrm{d}x$, where $f_n(x) = \begin{cases} 2x - 2x^2, & n = 1; \\ (\underbrace{f_1 \circ \cdots \circ f_1}_{n\ f_1\text{'s}})(x), & n \geq 2 \end{cases}$

Solution. Argue on the inverse function (which certainly exists). Let g be the inverse of f. By quadratic formula we have

$$g(x) = \frac{1 - \sqrt{1-2x}}{2}.$$

Now let $x = g_n(y)$. Since 0 and $\frac{1}{2}$ are fixed points of g, we have

$$\int_0^{\frac{1}{2}} f_n(x)\,\mathrm{d}x = \int_0^{\frac{1}{2}} y g_n'(y)\,\mathrm{d}y = [y g_n(y)]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} g_n(y)\,\mathrm{d}y = \frac{1}{4} - \int_0^{\frac{1}{2}} g_n(y)\,\mathrm{d}y.$$

Observe that the sequence $(g_n(y))_n$ is decreasing for any fixed $0 \leq y \leq \frac{1}{2}$. It is easy to check also that $0 \leq g_n(y) \leq \frac{1}{2}$ in this domain. Then $\lim_{n\to\infty} g_n(y) = \inf_{0\leq y\leq \frac{1}{2}} g_n(y) = 0$ since 0 is a fixed point of g. Then by dominated convergence we have

$$\lim_{n\to\infty} \int_0^{\frac{1}{2}} f_n(x)\,\mathrm{d}x = \frac{1}{4} - \int_0^{\frac{1}{2}} \lim_{n\to\infty} g_n(y)\,\mathrm{d}y = \frac{1}{4}.$$

:)

8. $\displaystyle\int_0^\infty \frac{x\cosh(x)}{\sinh(\pi x)}\,\mathrm{d}x$

Solution. Begin with evaluating the integral

$$\begin{aligned}
\int_0^\infty \frac{\sinh(ax)}{\sinh(\pi x)}\,\mathrm{d}x &= \int_0^\infty \frac{e^{-\pi x}(e^{ax} - e^{-ax})}{1 - e^{-2\pi x}}\,\mathrm{d}x \\
&= \sum_{n=0}^\infty \int_0^\infty e^{-(2n+1)\pi x}(e^{ax} - e^{-ax})\,\mathrm{d}x \\
&= \sum_{n=0}^\infty \left(\frac{1}{(2n+1)\pi - x} - \frac{1}{(2n+1)\pi + x} \right) \\
&= \frac{1}{2\pi} \sum_{n=0}^\infty \left(\frac{1}{n + \frac{1}{2} - \frac{x}{2\pi}} - \frac{1}{n + \frac{1}{2} + \frac{x}{2\pi}} \right) \\
&= \frac{1}{2\pi} \left(\sum_{n=0}^\infty \left(\frac{1}{n} - \frac{1}{n + \frac{1}{2} + \frac{x}{2\pi}} \right) - \sum_{n=0}^\infty \left(\frac{1}{n} + \frac{1}{n + \frac{1}{2} - \frac{x}{2\pi}} \right) \right) \\
&= \frac{1}{2\pi} \left(\psi\left(\frac{1}{2} + \frac{x}{2\pi} \right) - \psi\left(\frac{1}{2} - \frac{x}{2\pi} \right) \right) \\
&= \frac{\cot\left(\frac{\pi}{2} - \frac{x}{2}\right)}{2} \\
&= \frac{1}{2}\tan\left(\frac{x}{2}\right),
\end{aligned}$$

where we have used the series representation of the digamma function and the digamma reflection formula.

Now let $I(a) = \displaystyle\int_0^\infty \frac{\sinh(ax)}{\sinh(\pi x)}\,\mathrm{d}x$. Then we get

$$I'(a) = \int_0^\infty \frac{x\cosh(ax)}{\sinh(\pi x)}\,\mathrm{d}x = \frac{1}{4}\sec^2\left(\frac{x}{2}\right).$$

In particular $I'(1) = \dfrac{1}{4}\sec^2\left(\dfrac{1}{2}\right)$. :)

5 Set 2/Game 5

1. $\displaystyle\int_0^\infty \frac{\tan^{-1}(x)}{x^2+1}\,\mathrm{d}x$

   Solution.

$$\int_0^\infty \frac{\tan^{-1}(x)}{x^2+1}\,\mathrm{d}x = \frac{1}{2}[(\tan^{-1}(x))^2]_0^\infty = \frac{\pi^2}{8}.$$

   :)

2. $\displaystyle\int \frac{\sec^2(x)\csc^2(x)}{\tan^2(x)+\cot^2(x)+2}\,\mathrm{d}x$

   Solution.

$$\begin{aligned}\int \frac{\sec^2(x)\csc^2(x)}{\tan^2(x)+\cot^2(x)+2}\mathrm{d}x &= \int \frac{1}{\sin^4(x)+\cos^4(x)+\sin^2(x)\cos^2(x)}\mathrm{d}x\\ &= \int \frac{1}{\sin^4(x)+\cos^4(x)+2\sin^2(x)\cos^2(x)}\mathrm{d}x\\ &= \int 1\mathrm{d}x\\ &= x+C.\end{aligned}$$

   :)

3. $\displaystyle\int_0^1 \frac{1}{1+\sqrt{\frac{1-x+\sqrt{x}}{x+\sqrt{1-x}}}}\,\mathrm{d}x$

   Solution. By King's rule, we have

$$I := \int_0^1 \frac{1}{1+\sqrt{\frac{1-x+\sqrt{x}}{x+\sqrt{1-x}}}}\,\mathrm{d}x = \int_0^1 \frac{1}{1+\sqrt{\frac{x+\sqrt{1-x}}{1-x+\sqrt{x}}}}\,\mathrm{d}x = \int_0^1 \frac{\sqrt{\frac{1-x+\sqrt{x}}{x+\sqrt{1-x}}}}{\sqrt{\frac{1-x+\sqrt{x}}{x+\sqrt{1-x}}}+1}\,\mathrm{d}x.$$

   It follows $2I = \displaystyle\int_0^1 1\,\mathrm{d}x = 1$ and hence $I = \dfrac{1}{2}$. :)

4. $\displaystyle\int_0^1 \frac{1}{x} - \left\lfloor\frac{1}{x}\right\rfloor - \frac{1}{x^2}\left\lfloor\frac{1}{x}\right\rfloor^{-2}\,\mathrm{d}x$

Solution. Make the substitution $x = \dfrac{1}{y}$ and get

$$\begin{aligned}
\int_0^1 \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor - \frac{1}{x^2} \left\lfloor \frac{1}{x} \right\rfloor^{-2} \mathrm{d}x &= \int_1^\infty \frac{1}{y} - \frac{\lfloor y \rfloor}{y^2} - \frac{1}{\lfloor y \rfloor^2} \,\mathrm{d}y \\
&= \sum_{n=1}^{\infty} \int_n^{n+1} \frac{1}{y} - \frac{n}{y^2} - \frac{1}{n^2} \,\mathrm{d}y \\
&= \lim_{N\to\infty} \sum_{n=1}^{N} \left(\ln(n+1) - \ln(n) + \frac{n}{n+1} - \frac{n}{n} \right) - \frac{\pi^2}{6} \\
&= \lim_{N\to\infty} \sum_{n=1}^{N} \left(\ln(n+1) - \ln(n) + 1 - \frac{1}{n+1} - 1 \right) - \frac{\pi^2}{6} \\
&= \lim_{N\to\infty} \left(\ln(N+1) - \sum_{n=1}^{N} \frac{1}{n+1} \right) - \frac{\pi^2}{6} \\
&= \lim_{N\to\infty} \left(\ln(N+1) - \sum_{n=0}^{N} \frac{1}{n+1} \right) + 1 - \frac{\pi^2}{6} \\
&= 1 - \gamma - \frac{\pi^2}{6}.
\end{aligned}$$

:)

5. $\displaystyle\int_0^1 \sqrt{1 + \sqrt[3]{x-1}} \,\mathrm{d}x$

Solution. We use the inverse. Let $y = \sqrt{1 + \sqrt[3]{x-1}}$. Then it follows

$$x = (y^2 - 1)^3 + 1.$$

Further, $y(0) = 0$ and $y(1) = 1$. It follows that our integral is

$$1 - \int_0^1 (y^2-1)^3 + 1 \,\mathrm{d}y = 1 - \int_0^1 y^6 - 3y^4 + 3y^2 \,\mathrm{d}y = \frac{16}{35}.$$

:)

6. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(x^4+1)^2} \,\mathrm{d}x$

Solution. Using $y = x^4$ and beta function representation, we have

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{1}{(x^4+1)^2}\,\mathrm{d}x &= 2\int_0^{\infty} \frac{1}{(x^4+1)^2}\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^{\infty} \frac{y^{-\frac{3}{4}}}{(y+1)^2}\,\mathrm{d}x \\ &= \frac{1}{2}\,\mathrm{B}\left(\frac{1}{4},\frac{7}{4}\right) \\ &= \frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{7}{4}\right)}{2\Gamma(2)} \\ &= \frac{3\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)}{8} \\ &= \frac{3\cdot 2^{1-\frac{1}{2}}\sqrt{\pi}\Gamma\left(\frac{1}{2}\right)}{8} \\ &= \frac{3\sqrt{2}}{8}\pi.\end{aligned}$$

To evaluate the gamma function product, we have applied the Legendre duplication formula. :)

7. $\displaystyle\int_0^{2025} \min\left(\sqrt{\{x\}-\{x\}^2}, \left|\frac{\sqrt{3}\arcsin\left(\cos\left(\pi x\right)\right)}{\pi}\right|\right)\,\mathrm{d}x$

Solution. The key to understanding this problem lies in understanding what each function looks like. The first function is relatively easy: it is a collection of diameter 1 upper semicircles centered at $x = \dfrac{k}{2}$, $k \in 2\mathbb{Z}+1$.

The second function is somewhat harder to understand as-is, but break it down and it becomes relatively simple. First, we have $\sin^{-1}(\cos(\theta)) = \dfrac{\pi}{2} - \theta$, assuming all values involved are in the principal domain. Next, considering domains, $\sin^{-1}(\cos(\pi x))$ is periodic with period (of course) 2, being piecewise linear between points $\left(0, \dfrac{\pi}{2}\right)$, $\left(1, -\dfrac{\pi}{2}\right)$ and $\left(1, -\dfrac{\pi}{2}\right)$, $\left(2, \dfrac{\pi}{2}\right)$. It follows that $\left|\dfrac{\sqrt{3}\arcsin\left(\cos\left(\pi x\right)\right)}{\pi}\right|$ is a spike function consisting of isosceles triangles peaking at $\left(n, \dfrac{\sqrt{3}}{2}\right)$, $n \in \mathbb{Z}$.

[Insert some image here]

Combining the above, the required integral is a sum of sector areas. From n to $n+1$, the area is two sectors with angle $\tan^{-1}\left(\dfrac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right) = \dfrac{\pi}{3}$, i.e.

$$2\cdot\frac{\frac{\pi}{3}}{2\pi}\cdot\frac{\pi}{4} = \frac{\pi}{12}.$$

Thus the integral evaluates to $\dfrac{2025\pi}{12}$. :)

8. $\displaystyle\int \frac{\cos(x)+1-x^2}{(x\sin(x)+1)\sqrt{1-x^2}}\,\mathrm{d}x$

Solution. Let $y = \dfrac{x+\sin(x)}{x\sin(x)+1}$. Then it follows

$$\begin{aligned}
y' &= \frac{(x\sin(x)+1)(1+\cos(x)) - (x+\sin(x))(x\cos(x)+\sin(x))}{(x\sin(x)+1)^2} \\
&= \frac{x\sin(x) + x\sin(x)\cos(x) + 1 + \cos(x) - x^2\cos(x) - x\sin(x) - x\sin(x)\cos(x) - \sin^2(x)}{(x\sin(x)+1)^2} \\
&= \frac{\cos^2(x) + \cos(x) - x^2\cos(x)}{(x\sin(x)+1)^2} \\
&= \frac{(\cos(x) + 1 - x^2)\cos(x)}{(x\sin(x)+1)^2}.
\end{aligned}$$

It follows that

$$\frac{\cos(x)+1-x^2}{(x\sin(x)+1)\sqrt{1-x^2}}\,\mathrm{d}x = \frac{x\sin(x)+1}{\cos(x)\sqrt{1-x^2}}\,\mathrm{d}y.$$

Thus the trick is to simplify the RHS above. Passing cosine in and converting to sine we have

$$\begin{aligned}
\frac{x\sin(x)+1}{\sqrt{(1-\sin^2(x))(1-x^2)}} &= \frac{x\sin(x)+1}{\sqrt{1-\sin^2(x)-x^2+x^2\sin^2(x)}} \\
&= \frac{x\sin(x)+1}{\sqrt{(x\sin(x)+1)^2-(x+\sin(x))^2}} \\
&= \left(\frac{(x\sin(x)+1)^2-(x+\sin(x))^2}{(x\sin(x)+1)^2}\right)^{-\frac{1}{2}} \\
&= \frac{1}{\sqrt{1-y^2}}.
\end{aligned}$$

Hence our integral is just

$$\int \frac{1}{\sqrt{1-y^2}}\,\mathrm{d}y = \sin^{-1}\left(\frac{x+\sin(x)}{x\sin(x)+1}\right) + C.$$

:)